CO2 –AT02- EXPERMENTATION TOOLS

COURSE NAME : MOBILE COMPUTING NAME : PRADEEP M

COURSE CODE: ITA0329 REG NO : 192571050

Q2) Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment. Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.
- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design. Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

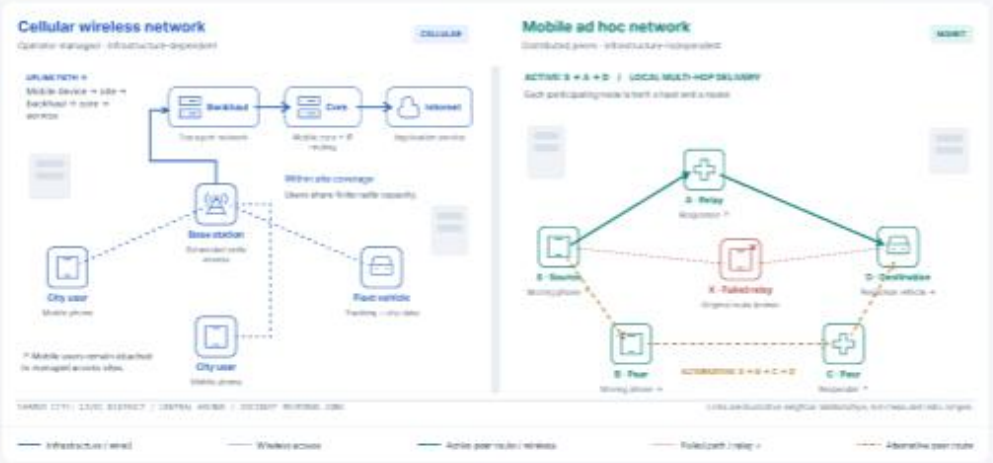
One smart city. Two connectivity models.

Comparative Network Architecture Design Using Figma

Cellular networks connect mobile users through managed infrastructure. Mobile ad hoc networks (MANETs) connect peers directly, with participating devices forwarding each other's traffic.

Dependency · Routing · Scale · Resilience · Mobility · Use

01 Two architectures, one shared urban environment



CELLULAR / Managed reach, fixed dependencies

Wide-area service and coordinated handover; access-site or backhaul outages can disrupt local service. Redundancy helps, but does not remove these dependencies.

MANET / Local autonomy, conditional resilience

No cellular site, core or gateway is required. Peers can route around a lost relay while connected alternatives exist; sparse or moving nodes can partition the network.

02 Choose for the operating context, not a universal winner

Managed wide-area connectivity

Use for citywide service and mobile access to external applications, where coverage and operator infrastructure are available.

Infrastructure-independent local collaboration

Use for temporary teams and local coordination when fixed access is unavailable, with enough connected peers to sustain routes.

COMPLEMENT, NOT REPLACE — Hybrid solutions can combine both models, but require explicit gateway and integration planning. Any gateway is optional for local MANET operation; it is not a prerequisite for peer communication.

03 Analyze the trade-offs

COMPARISON LENS	01 / Cellular wireless network	02 / Mobile ad hoc network
Infrastructure dependency	Requires base stations, backhaul and a mobile core. Operator deployment and maintenance enable managed coverage, but create fixed dependencies and significant cost.	No fixed access infrastructure is needed for local communication. Nodes provide forwarding themselves; usable links, sufficient peers and battery power are still required.
Routing mechanisms	Scheduled radio access connects users to a site; operator-managed transport and core routing carry traffic to services. Mobility management maintains sessions across cells.	Peers forward traffic over multiple radio hops. Reactive AODV discovers routes on demand; proactive OLSR maintains routes. Discovery, updates and repair add overhead.
Scalability	Cell reuse, scheduling and added sites support capacity growth. Finite spectrum, congestion, backhaul/core capacity and deployment cost remain practical constraints.	Shared radio airtime, interference, hop count and routing overhead constrain growth. More nodes can extend reach, but do not imply proportionally higher capacity.
Fault tolerance	Redundant sites, backhaul and core paths can improve resilience. Failure of local access infrastructure can still interrupt service; recovery depends on available alternatives.	Distributed paths can bypass a failed relay if connected alternatives exist. Losing key nodes or sparse deployment can cause partitions; route repair is not guaranteed.
Mobility support	Managed handover supports movement within network coverage. Cell transitions can introduce interruption; coverage gaps or overload can still cause loss of service.	Motion changes neighbor links and topology. Frequent changes trigger discovery or updates, with possible route breaks, packet loss, repair delay and battery consumption.
Application scenarios	Citywide voice and data, fleet tracking and connected public services—especially when broad coverage, managed capacity and external service access are priorities.	Disaster response after infrastructure damage, temporary field teams and local peer coordination; vehicular ad hoc use where suitable links and routing are available.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

Q3 · Routing protocol concept map

MOBILE COMPUTING / QUESTION 03

BL4 → ANALYZE Editable static concept map

Mobile wireless routing protocols & comparison map

DSDV vs AODV: proactive vs reactive routing behavior, condensed into a concise concept map.

CONCEPTUAL · NOT A SIMULATION

Packet paths explain protocol behavior. No simulated measurements or working prototype interactions are shown.

01 Mobile wireless routing protocols

Branch from the central topic into proactive and reactive routing, then compare discovery, maintenance, overhead, delay and mobility behavior.

CENTRAL TOPIC

Mobile wireless routing protocols

DSDV vs AODV: proactive vs reactive routing behavior

Compare how each protocol finds routes, handles breaks, consumes overhead and responds to mobility.

PROACTIVE / TABLE DRIVEN

DSDV

Destination-Sequenced Distance-Vector

Maintain routes continuously, whether or not they are used.

REACTIVE / ON DEMAND

AODV

Ad hoc On-demand Distance Vector

Find a route when it is needed; maintain active routes.

ROUTE DISCOVERY

DSDV	AODV
Routes are precomputed through neighbor table exchanges. A valid entry lets S forward without a discovery exchange.	No route is assumed initially. RREQ flooding and a returning RREP establish the route when traffic demands it.

ROUTE MAINTENANCE

DSDV	AODV
Invalidate affected entries and advertise unreachable metrics with newer sequence information. Triggered and subsequent table updates restore consistent routes.	Detect the next-hop failure, invalidate affected routes and notify upstream nodes with RERR. Rediscover at the source in this example; local repair is optional.

ROUTING OVERHEAD

DSDV	AODV
Periodic full dumps / incremental exchanges incur background cost even without application traffic. Mobility adds triggered update churn.	Avoids maintaining unused routes and often reduces idle overhead. RREQ floods, replies, errors and repeated repairs can still be expensive.

SETUP DELAY

DSDV	AODV
Low initial forwarding delay when the table entry is valid. Stale entries and convergence after movement can delay or disrupt delivery.	First-packet latency includes discovery and reply time. Active valid routes forward directly, but breaks introduce another discovery or repair delay.

HIGH MOBILITY

DSDV	AODV
Frequent changes can outpace propagation: stale tables, repeated invalidations and update churn reduce efficiency and can cause packet loss.	Active paths break more often, creating repeated discoveries, repair delays and control floods. Packets may be lost during disruption.

SUITABILITY CONCLUSION

Choose based on traffic and mobility

DSDV is a strong fit for low mobility and frequent traffic because ready routes reduce initial forwarding delay. AODV often fits better for moderate mobility and sparse/bursty traffic because it avoids maintaining unused routes. Neither universally wins under high mobility.